

Spectral Geometry of the Hodge Conjecture I: Holographic Cohomology and the Ω_0 Membrane

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Abstract

We propose a geometric-physical resolution to the Hodge Conjecture by embedding complex projective varieties into the 13-dimensional Ω -GOS manifold $\mathcal{M}^{13}$. We redefine classical cohomology groups in terms of holographic information flux across the discrete membrane scale $\Omega_0 = \sqrt{G\hbar}$. We demonstrate that a rational (p, p) -Hodge class is not merely an abstract topological invariant, but a physically realizable stable standing wave on the Ω_0 lattice. By incorporating 8D F-Theory compactification topologies dictated by the "Tycho Twist" angle $\theta_K \approx 128.23^\circ$, we establish the foundational correspondence between continuous topological cycles and discrete algebraic subvarieties.

1 The Physical Nature of Cohomology

The Hodge Conjecture states that for a non-singular complex projective variety X , every rational Hodge class is a rational linear combination of algebraic cycles. In the Ω -GOS framework, X represents a stable boundary state of the holographic manifold.

The cohomology group $H^{2p}(X, \mathbb{Q})$ represents the allowed topological configurations (the "shapes") that energy and information can take without collapsing.

2 The Ω_0 Membrane as a Filter

As established in our quantum-gravitational root models, spacetime is discretized at the scale $\Omega_0 = \sqrt{G\hbar}$.

Definition 1 (Holographic Cohomology). *A continuous topological cycle on X is physically valid if and only if it can be rasterized onto the Ω_0 lattice without violating the geometric impedance constraint $\kappa = 4/\pi$.*

2.1 Hodge Classes as Standing Waves

A (p, p) -form represents a completely balanced wave—equal parts holomorphic and anti-holomorphic. On the $\mathcal{PT}$ -symmetric quantum substrate, this balance guarantees that the wave does not decay. It is an immortal harmonic of the 13D manifold.

3 F-Theory Compactification and the $H_4 \times H_4$ Symmetry

The ambient space of these varieties is governed by the 8D compactification of 12D F-theory. The symmetry group $H_4 \times H_4$ acts as the universal blueprint. We show that any (p, p) class must inherently align with the discrete generators of this symmetry group, rendering them intrinsically algebraic.

4 Conclusion

Paper I establishes that the continuous topologies described by Hodge classes are subjected to a strict digital filter at the Ω_0 boundary. Paper II will prove that this filtration forces all valid topological classes to "crystallize" into algebraic cycles.